

Deep Reinforcement Learning of Extra-synaptic Control in *C. elegans* Chemosensory Response

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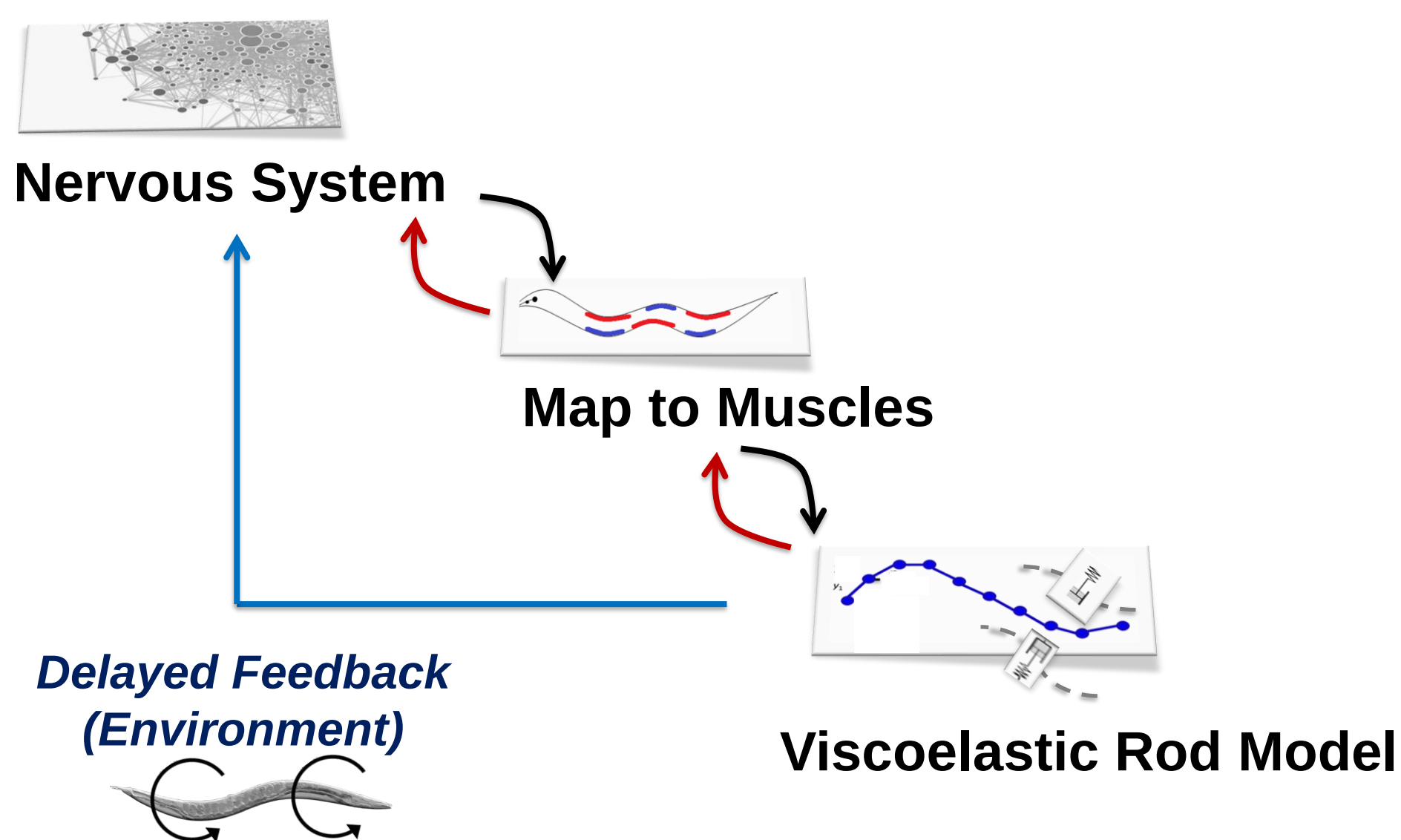
Introduction

We propose to infer **Extra-synaptic Interactions** through a combination of **Deep Q-Network (DQN)** learning along with ***C. elegans* Neuro-mechanical Model** simulating chemosensory responses.

Key Points:

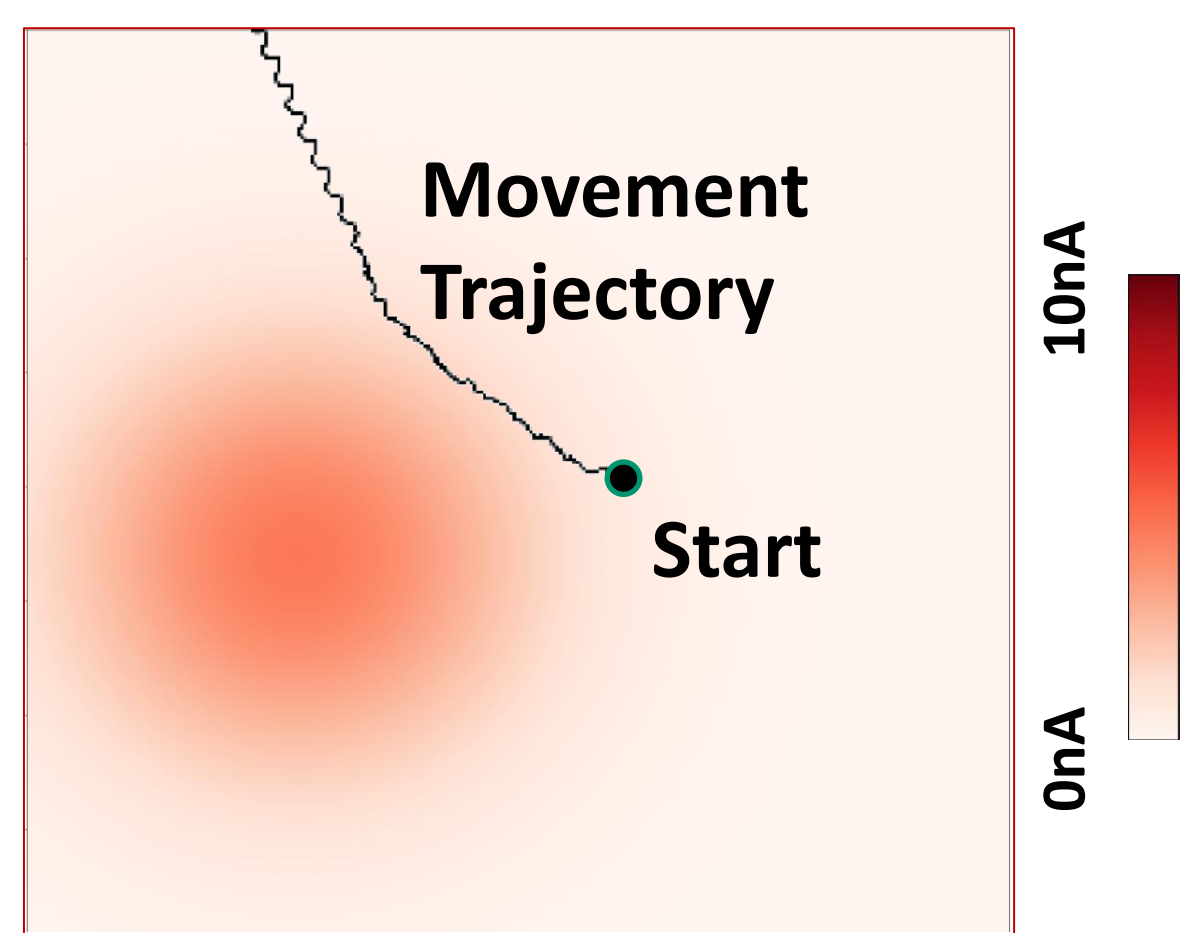
- C. elegans* model** generates **aversive** chemosensory responses to **CO₂** and **O₂** like *in-vivo* experiments.
- We propose **RL-DQN method** to seek for **extra-synaptic currents** which **modulate** the responses.
- RL-DQN** infers **plausible extra-synaptic** controls in both CO₂ and O₂ responses which switch response from **aversion** to **attraction**.
- RL-DQN** modulated network dynamics result in **activity consistent** with the **activity measured in-vivo**.

Neuro-mechanical Model of *C. elegans*

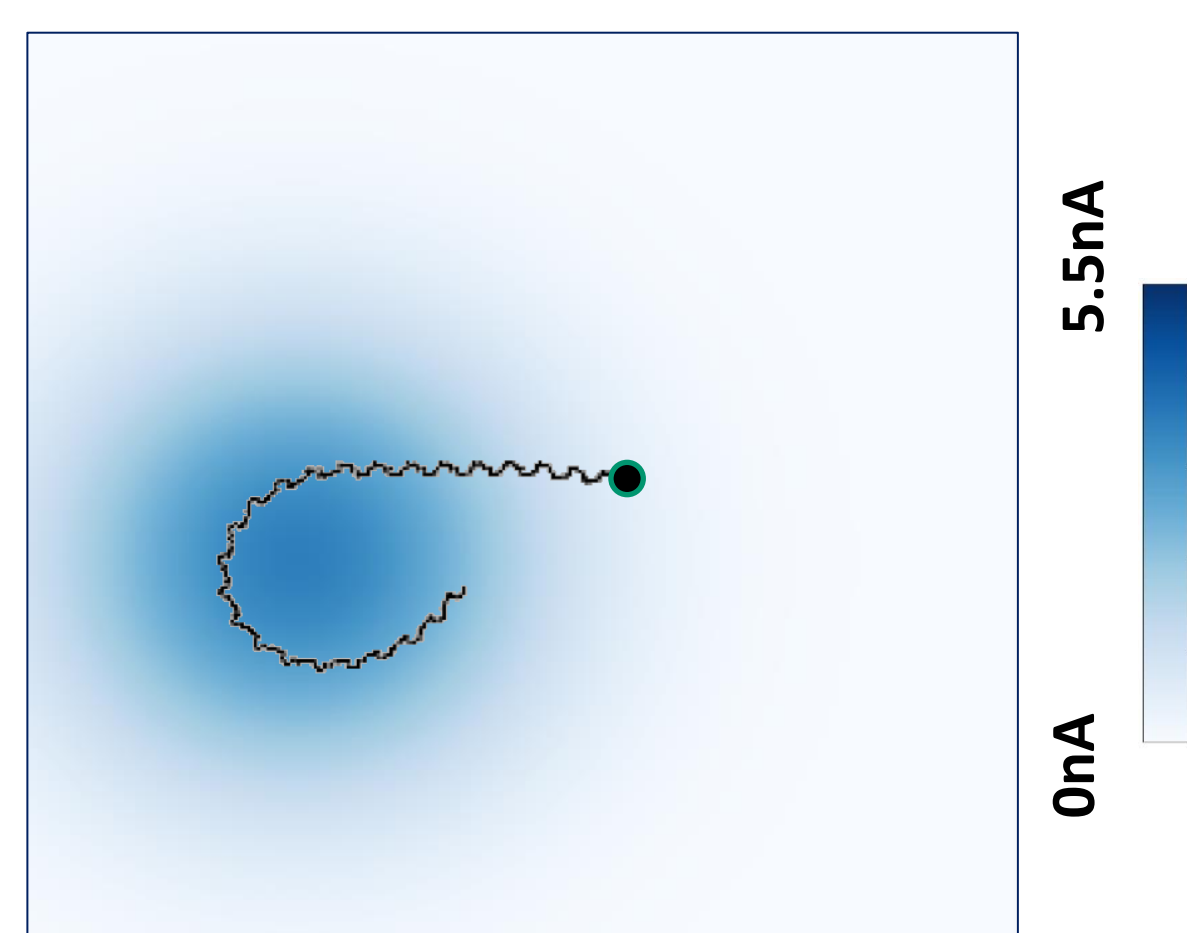


Simulating *C. elegans* Response to O₂ and Co₂

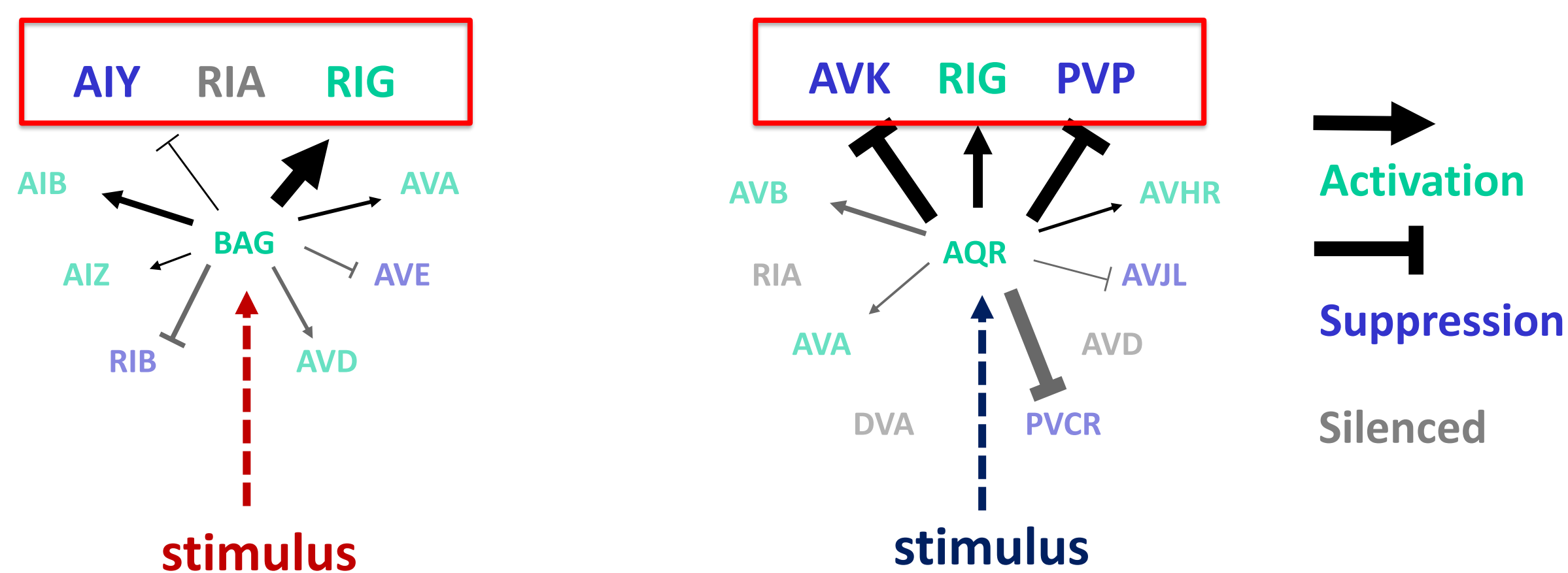
In silico 'Wild' CO₂
Response to BAG gradient



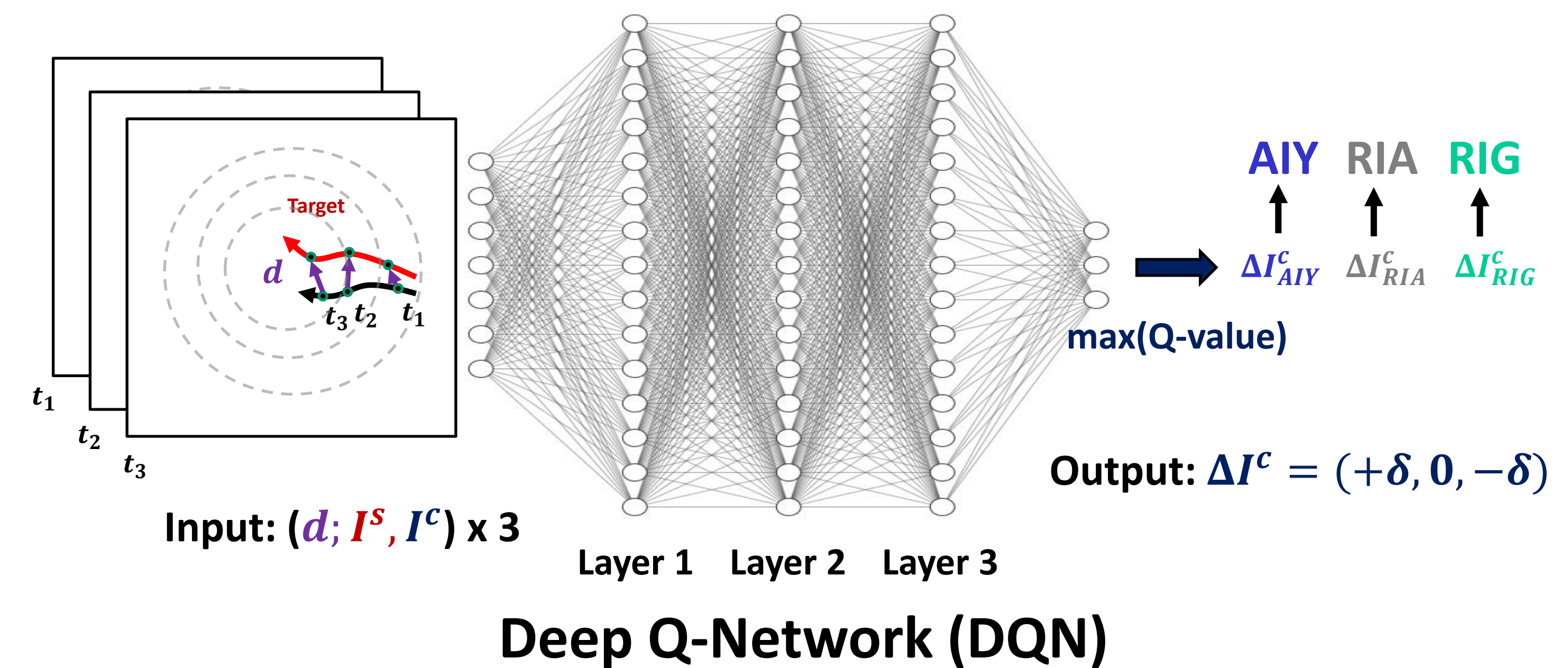
In silico 'Wild' O₂
Response to AQR gradient



Neighbor Neurons Activity (Voltage)

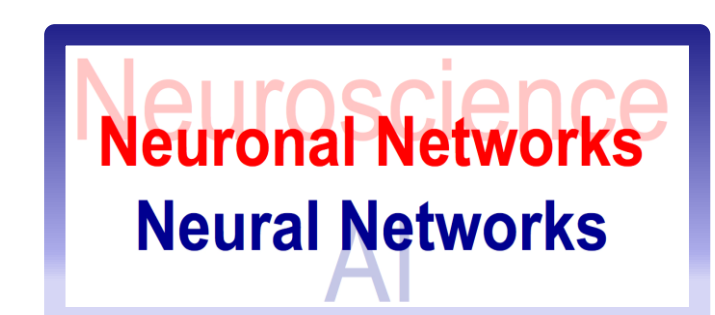
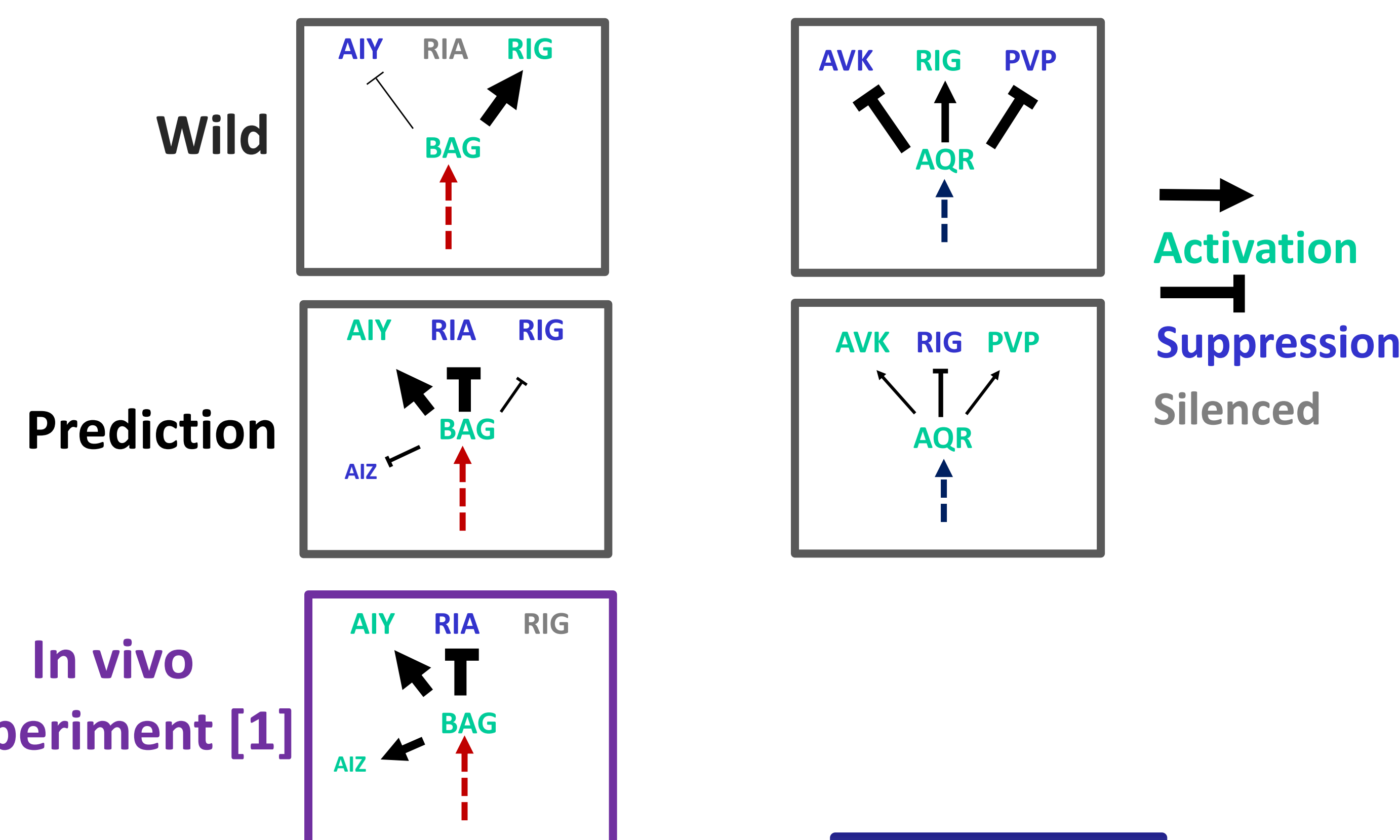
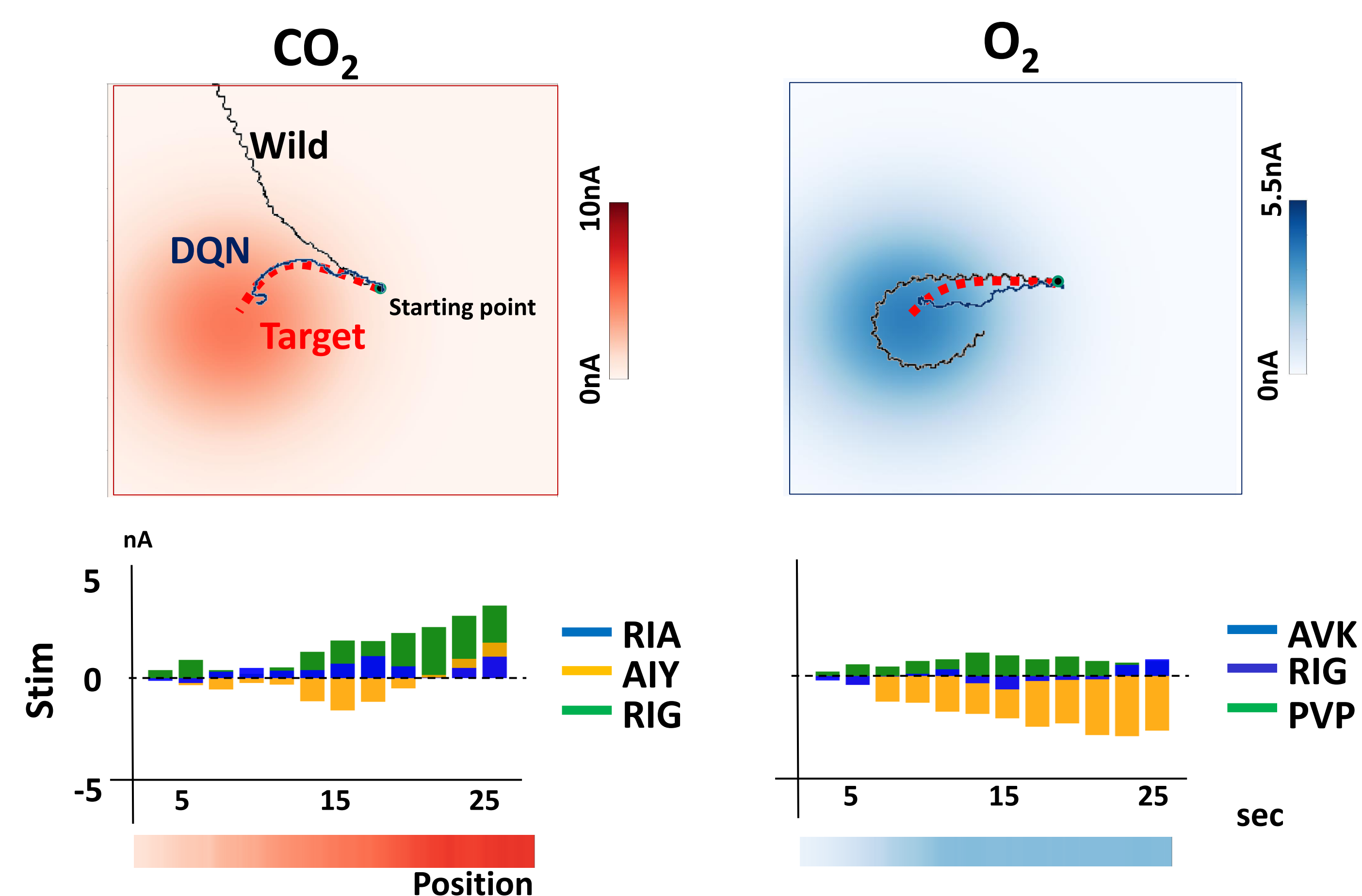


DQN Architecture to control Extra-synaptic Current Modulation



Target and DQN Modulated Trajectories

DQN inferred modulated Response



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References

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